

Computing MLEs

- **Maximum Likelihood Estimator**(mle): we say that $\hat{\theta} = \hat{\theta}(\mathbf{X})$ is a maximum likelihood estimator if

$$\hat{\theta} = \arg \max_{\theta} L(\theta; \mathbf{X}).$$

- Solve the equation

$$\frac{\partial l(\theta)}{\partial \theta} = 0$$

to obtain the mle. This equation is called **Estimation Equation**(EE).

- When L or ℓ is not differentiable, try to draw a picture of the likelihood function.
- Is the MLE unique? No!

- Exponential Distribution: Suppose the common pdf is $\text{exponential}(\theta)$, which is given by

$$f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}}, \quad x > 0.$$

The log likelihood function is

$$l(\theta) = -n \log \theta - \theta \sum_{i=1}^n x_i.$$

$$\frac{\partial l(\theta)}{\partial \theta} = -n\theta^{-1} + \theta^2 \sum_{i=1}^n x_i$$

So, $\hat{\theta} = \bar{X}$ is the mle of θ .

- Laplace Distribution: Let $X_1, \dots, X_n$ be i.i.d. with density

$$f(x; \theta) = \frac{1}{2} e^{-|x-\theta|}, -\infty < x < \infty, -\infty < \theta < \infty.$$

This pdf is also called *Double Exponential Distribution*. The log likelihood is

$$l(\theta) = -n \log 2 - \sum_{i=1}^n |x_i - \theta|.$$

$$\frac{\partial l(\theta)}{\partial \theta} = \sum_{i=1}^n \text{sgn}(x_i - \theta),$$

where $\text{sgn}(t) = 1, 0, -1$ depending on whether $t > 0, = 0, < 0$. So the mle is the median of $\{x_1, \dots, x_n\}$.

- Logistic Distribution: Let $X_1, \dots, X_n$ be i.i.d. with pdf

$$f(x; \theta) = \frac{\exp(-(x - \theta))}{\{1 + \exp(-(x - \theta))\}^2}, -\infty < x < \infty, -\infty < \theta < \infty.$$

The log likelihood

$$l(\theta) = n\theta - n\bar{x} - 2 \sum_{i=1}^n \log(1 + \exp\{-(x_i - \theta)\}).$$

$$l'(\theta) = n - 2 \sum_{i=1}^n \frac{\exp(-(x_i - \theta))}{1 + \exp(-(x_i - \theta))}$$

- Uniform Distribution: Let $X_1, \dots, X_n$ be i.i.d. with $\text{uniform}(0, \theta)$, i.e., $f(x) = 1/\theta$, $x \in (0, \theta]$. The likelihood is

$$L(\theta) = \theta^{-n} I_{[\max\{X_i\}, \infty)}(\theta)$$

for all $\theta > 0$, where I is the indicator function. Hence, mle is $\hat{\theta} = \max\{X_i\}$.

- Let $X_1, \dots, X_n$ be from $\text{Uniform}((\theta - \frac{1}{2}, \theta + \frac{1}{2}))$. What is the MLE of θ ?

